THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET C

Chapters 11 & 12: **The World of Algorithms, Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If Step 5 (writing the final carry) is skipped, then 856 + 479 would be written as: 1

(a) 3350 **(b)** 1235 **(c)** 1335 **(d)** 335

2. A list, which is used to store divisors in order, is an example of a: 1

(a) variable **(b)** fraction **(c)** theorem **(d)** data structure

3. The word ‘algorithm’ is derived from the name of: 1

(a) Al-Khwārizmī **(b)** Āryabhaṭa **(c)** Pythagoras **(d)** Euclid

4. **Assertion (A):** In the addition algorithm, the carry is never more than 1. 1

Reason (R): The sum of two digits and a carry of at most 1 is at most 19.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

5. While adding 268 and 175 by the column algorithm, the number of columns that produce a carry of 1 is: 1

(a) 1 **(b)** 3 **(c)** 4 **(d)** 2

6. The diagonals of a convex quadrilateral: 1

(a) are always perpendicular **(b)** never intersect

(c) are always equal **(d)** intersect each other

7. The converse of ‘the opposite angles of a parallelogram are equal’ is: 1

(a) all quadrilaterals have equal opposite angles

(b) if the opposite angles of a quadrilateral are equal, it is a parallelogram

(c) if a quadrilateral has equal angles, it is a rectangle

(d) if a quadrilateral is a rectangle, opposite angles are equal

8. The three medians of a triangle meet at a point called the: 1

(a) orthocentre **(b)** circumcentre **(c)** incentre **(d)** centroid

9. For tiling the plane, the angles meeting at a point of the tiling must add up to: 1

(a) 720° **(b)** 360° **(c)** 180° **(d)** 90°

10. **Assertion (A):** The centroid of a triangle divides every median in the ratio 2 : 1. 1

Reason (R): The medians of a triangle are concurrent.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does gcd(99, 2) need many steps in Euclid's subtraction algorithm but only two in the division algorithm? 2

12. Write a short note on the origin of the word ‘algorithm’. 2

13. Name two real-life devices that use parallelogram properties. 2

14. In $\triangle ABC$, the line through the midpoint P of AB parallel to BC meets AC at Q. If AC $= 18$ cm, find AQ and QC. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. **Case study:** A student adds 4856 and 3779 on paper. She writes 8625 as the answer. **3**
(i) Find the correct sum.
(ii) In which column did she fail to add the carry?
(iii) Which step of the algorithm did she ignore?

16. **Case study:** A parallelogram linkage has rods AB $= 40$ cm and BC $= 25$ cm with $\angle ABC = 110^\circ$. **3**
(i) Find $\angle BCD$.
(ii) Find the perimeter of the linkage.
(iii) If the rods are pushed so that $\angle ABC = 70^\circ$, what stays the same?

17. **Case study:** Rohan lists divisors(50) = [1, 2, 5, 10, 25, 50] and divisors(75) = [1, 3, 5, 15, 25, 75]. **3**
(i) Write the list of common divisors.
(ii) Find gcd(50, 75).
(iii) Why is the gcd the rightmost element?

OR

Case study: A delivery company packs 144 apples and 108 oranges into identical boxes with no fruit left over. **3**
(i) Which gcd gives the largest number of fruits possible in each box if boxes hold only one fruit type?
(ii) Compute it by the division algorithm.
(iii) How many boxes of each fruit?

18. **Case study:** To find the width of a pond, a surveyor marks A on one side and B and C on the opposite banks, then takes P and Q as the midpoints of AB and AC on land. PQ $= 35$ m. **3**
(i) State the theorem used.
(ii) Find BC.
(iii) How is PQ related to BC in direction?

OR

Case study: A field ABCD has diagonals AC $= 60$ m and BD $= 80$ m. Farmers join the midpoints P, Q, R, S of the sides to make a pond. **3**
(i) Name the shape PQRS.
(ii) Find its perimeter.
(iii) If AC and BD are perpendicular, what is PQRS?

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A school wants to seat 84 boys and 126 girls in rows so that each row has only boys or only girls and every row has the same number of students. **5**
(i) What is the largest row size?
(ii) Show the divisor lists' common part.
(iii) How many rows of boys and girls result?
(iv) Which algorithm would you use for 8400 and 12600 and why?

OR

Case study: Ishaan wants to know how the work in the column-addition algorithm grows with the number of digits. **5**
(i) How many single-digit additions does one column need at most?
(ii) For two n-digit numbers, how many columns?
(iii) For n = 4 and 15 s per column, find the time.
(iv) By counting dots at 1 s per dot, estimate the time for 4-digit numbers whose sum is about 6000 and compare.

20. **Case study:** A farmer's field ABCD has diagonals AC $= 70$ m and BD $= 70$ m which meet at right angles. A fence is built joining the midpoints P, Q, R, S of the sides in order. **5**
(i) Name PQRS and justify.
(ii) Find each side.
(iii) Find the perimeter and the area of PQRS.
(iv) What is the area of the field?

OR

Case study: A triangular garden ABC has medians AD, BE and CF meeting at G. AD $= 27$ m, BE $= 36$ m. **5**
(i) State the ratio in which G divides each median.
(ii) Find AG, GD, BG, GE.
(iii) If the third median CF $= 30$ m, find CG and GF.
(iv) Why are the three medians guaranteed to meet at one point?

END OF QUESTION PAPER — Total: 40 Marks

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